The CTC Algorithm — Solutions Manual

manim-concepts

Chapter 1

Counting rules

Practice problems

Problem 1.1. A per-frame labeller writes one of 3 symbols at each of 4 frames, independently. How many distinct label sequences can it write?

Solution. Four independent stages of 3 options each: $3 \times 3 \times 3 \times 3 = 3^4 = 81$. (This is the raw path space the next chapter's model writes over — the number returns almost immediately.)

Problem 1.2. Eight runners; how many distinct podiums (gold, silver, bronze) are possible?

Solution. $P_3^8 = 8 \times 7 \times 6 = 336$. In factorial form, $8!/5! = 336$: write the full ordering count and cancel the five unused slots' tail.

Problem 1.3. From the same eight runners, how many three-person relay *teams* (no roles) can be formed? Check your answer against the previous problem using the overcount argument.

Solution. Each unordered team of 3 appears $3! = 6$ times among the 336 podiums, so $\binom{8}{3} = 336/6 = 56$. Equivalently $8!/(3!\,5!) = 56$. The check is the definition: $C = P/r!$.

Problem 1.4. How many distinct arrangements does the word SEEDED have?

Solution. Six letters with repeats $E \times 3$, $D \times 2$, $S \times 1$:

$$\frac{6!}{3!\,2!\,1!} = \frac{720}{12} = 60.$$

The partition rule — the divide-out move applied per block of equal letters.

Problem 1.5. [stretch] A walker on a grid moves from the bottom-left corner to the top-right corner of a 2-row-by-4-column lattice of steps, moving only right or up — 4 rights and 2 ups in some order. How many routes are there?

Solution. A route is an arrangement of the multiset $\{R, R, R, R, U, U\}$: $\binom{6}{2} = \frac{6!}{4!\,2!} = 15$ routes — choosing which 2 of the 6 steps go up. Keep this number; a lattice with exactly this count is about to appear wearing a completely different costume.

Chapter 2

The alignment problem, and CTC's answer

Practice problems

Problem 2.1. Apply the collapse map: what are $\mathcal{B}(\text{AA}\varepsilon\text{B})$, $\mathcal{B}(\varepsilon\text{ABB})$, and $\mathcal{B}(\text{ABBA})$?

Solution. $\mathcal{B}(\text{AA}\varepsilon\text{B})$: merge the AA, drop the blank $\to$ AB. $\mathcal{B}(\varepsilon\text{ABB})$: nothing adjacent merges except BB, drop the blank $\to$ AB. $\mathcal{B}(\text{ABBA})$: merge BB $\to$ ABA — the trailing A is a *new* A, separated by the B, so it survives: ABA, not AB.

Problem 2.2. Show by example that the collapse order matters: find a path that yields a double letter under merge-then-drop but loses it under drop-then-merge.

Solution. Take LεL. Merge-then-drop: no adjacent repeats to merge, drop the blank $\to$ LL — the double letter survives. Drop-then-merge: dropping the blank first gives LL adjacent, which then merges $\to$ L. Only merge-then-drop lets the blank protect a genuine double letter, which is why the map is defined in that order.

Problem 2.3. Over $T = 3$ frames with alphabet $\{A, \varepsilon\}$, how many raw paths are there, and how many collapse to the transcript "A"?

Solution. Raw: $2^3 = 8$. Collapsing to A means the path shows one contiguous block of A's and blanks elsewhere: the block can occupy frames $\{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}$ — 6 paths. The trellis agrees: unit-weight columns on the 3 states $(\varepsilon, A, \varepsilon)$ run $(1,1,0) \to (1,2,1) \to (1,3,3)$, and the two accepting states give $3+3=6$.

Problem 2.4. Using the repeat-free count $\binom{T+U}{T-U}$, how many paths spell a 2-letter transcript with no repeated letter at $T=5$?

Solution. $\binom{5+2}{5-2} = \binom{7}{3} = 35$ paths — the count 35 the road meets again when blank dominance is tallied at $T=5$.

Problem 2.5. [stretch] The chapter claims greedy decoding can be wrong. Over the two-symbol alphabet $\{A, \varepsilon\}$ at $T=2$, construct per-frame scores where the frame-wise argmax path collapses to a transcript that is NOT the highest-probability transcript.

Solution. Score both frames identically: $y_t(A) = 0.4$, $y_t(\varepsilon) = 0.6$. Greedy takes the blank at each frame — path $\varepsilon\varepsilon$ — and decodes the *empty* transcript, whose only path has mass $0.6 \times 0.6 = 0.36$. The transcript "A" pools three paths: $\mathcal{B}^{-1}(\mathrm{A}) = \{\mathrm{AA},\ \mathrm{A}\varepsilon,\ \varepsilon\mathrm{A}\}$ with mass $0.16 + 0.24 + 0.24 = 0.64$. (The two transcripts exhaust the outcomes: $0.36 + 0.64 = 1$.) Greedy answers "nothing was said"; the sum answers "A", by nearly two to one — frame-wise favourites lose to a transcript that spreads its weight across many paths, which is exactly why the loss is built on the sum.

Chapter 3

Dynamic programming

Practice problems

Problem 3.1. Computing F_{10} by the bare recurrence $F_n = F_{n-1} + F_{n-2}$ (with $F_0 = 0$, $F_1 = 1$ answered directly), how many function calls are made in total? How many *distinct* questions were asked?

Solution. The call count obeys $c_n = 1 + c_{n-1} + c_{n-2}$, which solves to $c_n = 2F_{n+1} - 1$: the tree grows at the sequence's own rate. For $n = 10$: $c_{10} = 2F_{11} - 1 = 2 \cdot 89 - 1 = 177$ calls — to answer the 11 distinct questions F_0 through F_{10}. The memo table answers each distinct question once: eleven entries, one addition each.

Problem 3.2. Label every node of the 4-right, 2-up lattice with the number of routes reaching it, using only the addition rule. Confirm the far corner agrees with the counting chapter's $\binom{6}{2}$.

Solution. Filling row by row (bottom row first):

$$\begin{array}{ccccc} 1 & 3 & 6 & 10 & 15 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 & 1 \end{array}$$

The corner holds $15 = \binom{6}{2}$, matching the choose-the-ups count — and matching 15, the AB path count, a third meeting with the same number: the trellis *is* a lattice walk with different legal steps.

Problem 3.3. Extend the mini trellis one more frame: for transcript A over $T = 4$ frames, compute the fourth column from $(1, 3, 3)$ and read off the accepted count. Then confirm the count by a direct argument about which raw paths collapse to A.

Solution. Each cell adds its own state's previous count and the one above: $(1,\ 3{+}1,\ 3{+}3) = (1,4,6)$, and the two accepting states give $4+6=10$. Directly: a path collapsing to A is exactly one contiguous block of A-frames inside 4 frames — intervals of length $1,2,3,4$ number $4{+}3{+}2{+}1=10$. Two routes, one answer, and the recurrence wrote three cells to the enumeration's sixteen paths.

Problem 3.4. [stretch] At $T=100$ frames and $U=50$ letters, compare the work the two routes perform: how many cells does the trellis fill, and how does that stand against the raw repeat-free path count? What would happen to the table if a path's legal continuations depended on its last *two* states instead of one?

Solution. The trellis fills $T(2U+1) = 100 \times 101 =$ 10,100 cells, a handful of operations each, against roughly 2×10^{40} paths — about 36 orders of magnitude apart. If legality read the last two states, "where the path stands" would no longer determine its future: cells would need to be *pairs* of states, squaring the state count to order 10^4 and the table to order 10^6 — still polynomial, but the lesson generalises: dynamic programming pays exactly for how much past the future needs. When no small summary exists, the table, not the path list, becomes the exponential object.

Chapter 4

Independence

Practice problems

Problem 4.1. Roll two fair dice. Let $A = \{\text{first die shows 5 or 6}\}$ and $B = \{\text{second die shows 3 or less}\}$. Compute $P(A \cap B)$ and check it against the product rule.

Solution. The intersection is a rectangle of $2 \times 3 = 6$ cells out of 36: $P(A \cap B) = \frac{6}{36} = \frac{1}{6}$. The margins give $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{2}$, and $\frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$ — the product rule holds, because an event about the first die and an event about the second always cut the grid as perpendicular bands, and a rectangle's share of the grid is width times height.

Problem 4.2. On one fair die, test $A = \{\text{even}\}$ for independence against $B = \{1, 2, 3, 4\}$ and against $B' = \{1, 2, 3\}$.

Solution. Against B: $A \cap B = \{2, 4\}$, so $P(A \cap B) = \frac{1}{3} = \frac{1}{2} \cdot \frac{2}{3}$ — independent, inside a single roll. Against B': $A \cap B' = \{2\}$, so $P(A \cap B') = \frac{1}{6}$ while $P(A)P(B') = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$ — dependent. Moving one pip from B to B' flipped the verdict: independence is arithmetic about the measure, not intuition about mechanisms.

Problem 4.3. Show that $A = \{\text{even}\}$ and $B = \{1, 3, 5\}$ on a fair die are not independent, and say how much information seeing B carries about A.

Solution. $A \cap B = \emptyset$, so $P(A \cap B) = 0$, while $P(A)P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. The product test fails as loudly as it can: knowing B occurred settles A completely (it did not happen). Mutually exclusive events with positive probability are never independent — they are maximally dependent.

Problem 4.4. Flip two fair coins. Let $A = \{\text{first is H}\}$, $B = \{\text{second is H}\}$, $C = \{\text{exactly one head}\}$. Verify that every pair is independent, and that the three events are not mutually independent.

Solution. $P(A) = P(B) = P(C) = \frac{1}{2}$. Pairs: $A \cap B = \{\text{HH}\}$, $A \cap C = \{\text{HT}\}$, $B \cap C = \{\text{TH}\}$ — each of probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$, so all three pairs factor. But $A \cap B \cap C$ demands two heads *and* exactly one head: impossible, so its probability is $0 \neq \frac{1}{8}$. Pairwise independence does not give mutual independence — which is why the chain formula's condition says *mutual*.

Problem 4.5. [stretch] Draw two cards from a shuffled deck. Compute $P(\text{both aces})$ with replacement and without, and — without replacement — $P(\text{second card is an ace})$ before anyone looks at the first.

Solution. With replacement the draws are independent trials: $\left(\frac{1}{13}\right)^2 = 1/169$. Without, the pool shrinks: $\frac{4}{52} \cdot \frac{3}{51} = 1/221$ — smaller, because the first ace depletes the aces. Yet $P(\text{second is ace}) = \frac{1}{13}$ exactly (204/2652 enumerated): as long as no one looks at the first card, the second is just a uniformly random card. Depletion changes the *joint* behaviour, not the margin — which is exactly what "dependent" means.

Chapter 5

Conditional probability

Practice problems

Problem 5.1. Three fair coins are flipped. What is the probability all three landed heads — before any news, and then given that the first coin landed heads?

Solution. Eight equally likely outcomes; HHH is one: $\frac{1}{8}$. Given the first is heads, four outcomes survive (HHH, HHT, HTH, HTT) and HHH is one of them: $\frac{1}{4}$. The definition agrees: $P(\text{HHH} \mid \text{first H}) = \frac{1/8}{1/2} = \frac{1}{4}$ — restriction then renormalization, in numbers.

Problem 5.2. Two cards are drawn without replacement. Use the multiplication rule to price $P(\text{both aces})$, identifying each factor as a probability or a conditional probability.

Solution. $P(A_1) = \frac{4}{52}$. Given the first card was an ace, the second draw faces 51 cards of which 3 are aces: $P(A_2 \mid A_1) = \frac{3}{51}$. So

$$P(A_1 \cap A_2) = \frac{4}{52} \cdot \frac{3}{51} = 1/221.$$

The independence chapter asserted this number with no license shown; the license is the multiplication rule — the shrinking pool is a conditional probability.

Problem 5.3. An urn holds 5 red and 2 blue balls; two are drawn without replacement. Compute $P(\text{second is red})$ by total probability over the first draw, and compare it with $P(\text{first is red})$.

Solution.

$$P(R_2) = P(R_1)\,P(R_2 \mid R_1) + P(B_1)\,P(R_2 \mid B_1) = \frac{5}{7} \cdot \frac{4}{6} + \frac{2}{7} \cdot \frac{5}{6} = \frac{20+10}{42} = \frac{5}{7}.$$

Exactly $P(R_1)$. Before anyone looks at the first ball, the second draw is just a uniformly random ball — the same margin-preserving symmetry the aces showed. Dependence lives in the joint behaviour; total probability averages it back out.

Problem 5.4. A test catches 9 in 10 sick people and falsely flags 1 in 10 healthy people. Compute $P(\text{sick} \mid +)$ in whole-person counts for a 100-person cohort at 10% prevalence, and for a 1000-person cohort at 1% prevalence.

Solution. At 10%: 10 sick yield 9 positives; 90 healthy yield 9 — $P(\text{sick} \mid +) = 9/18 = \frac{1}{2}$. At 1%: 10 sick yield 9 positives; 990 healthy yield 99 — $P(\text{sick} \mid +) = 9/108 = 1/12$. The test never changed; the prior did, and it rode inside the counts. The definition route ($P(\text{sick})P(+ \mid \text{sick})/P(+)$, with the denominator by total probability) lands on the same two fractions.

Problem 5.5. [stretch] One of two coins is chosen by a fair flip: one lands heads with probability $\frac{9}{10}$, the other $\frac{1}{10}$. The chosen coin is flipped twice. Show the flips are dependent, but conditionally independent given the coin.

Solution. Each flip's margin is $P(H) = \frac{1}{2}(\frac{9}{10} + \frac{1}{10}) = \frac{1}{2}$. Jointly,

$$P(H_1 \cap H_2) = \frac{1}{2}\left(\frac{9}{10}\right)^2 + \frac{1}{2}\left(\frac{1}{10}\right)^2 = 41/100 \;\neq\; \frac{1}{4} = P(H_1)P(H_2)$$

— dependent: a first head is evidence you hold the heads-heavy coin. But given the coin, the flips factor exactly: $P(H_1 \cap H_2 \mid \text{coin}) = p^2$ with $p = \frac{9}{10}$ or $\frac{1}{10}$. The coin is the common cause; conditioning on it switches the dependence off — which is the exact shape of CTC's "independent given the input".

Chapter 6

Bayes' rule

Practice problems

Problem 6.1. In the Diseasitis cohort — 100 students, 20 sick of whom the test flags 9 in 10, 80 healthy of whom it falsely flags 3 in 10 — a student tests positive. Compute the probability they are sick using whole-person counts only.

Solution. Sick positives: $\frac{9}{10} \cdot 20 = 18$. Healthy positives: $\frac{3}{10} \cdot 80 = 24$. The positives total 42, and the share of them that are sick is $18/42 = 3/7$. No formula was used — the prior (20 sick in 100) rode inside the counts.

Problem 6.2. Recompute the Diseasitis posterior in the odds form, and check it against the counts.

Solution. Prior odds: 20:80 = 1:4. Likelihood ratio: $\frac{9/10}{3/10} = 3$. Posterior odds: $1{:}4 \times 3 = 3 : 4$ — literally the 18:24 of the counted cohort — and converting odds to probability, $\frac{3}{3+4} = 3/7$. Two routes, one answer; the waterfall is the cohort with renormalization deferred.

Problem 6.3. A test has sensitivity $\frac{9}{10}$ and false positive rate $\frac{1}{10}$. Using the odds form, find $P(\text{sick} \mid +)$ for a patient from a population at 10% prevalence, and for one at 1%.

Solution. The likelihood ratio is $\frac{9/10}{1/10} = 9$. At 10% prevalence the prior odds are 1:9, so posterior odds $1{:}9 \times 9 = 1{:}1$, i.e. $\frac{1}{2}$. At 1% the prior odds are 1:99, so $1{:}99 \times 9 = 9{:}99 = 1 : 11$, i.e. $1/12$. These reproduce the conditional chapter's counted 9/18 and 9/108 exactly — the test supplied one number, and each patient supplied their own prior.

Problem 6.4. One of two coins (heads $\frac{9}{10}$ versus $\frac{1}{10}$) is chosen by a fair flip. Track the odds that you hold the heads-heavy coin after observing H, after HH, and after HT — and give each as a probability.

Solution. Each head multiplies the odds by $\frac{9/10}{1/10} = 9$; each tail by $\frac{1/10}{9/10} = \frac{1}{9}$. From 1:1: after H, 9:1, probability $\frac{9}{10}$; after HH, $81 : 1$, probability $\frac{81}{82}$; after HT, $9 \times \frac{1}{9} = 1$, exactly 1:1 — probability $\frac{1}{2}$, as if nothing had been seen. Full conditioning over both coins and all flip sequences gives the same three posteriors, which is the licensed multiplication at work: the flips are conditionally independent given the coin.

Problem 6.5. [stretch] You pick door 1; the host opens door 3, showing a goat. Compute the probability that switching to door 2 wins under each protocol: (a) the standard host — never opens your door or the car's, coin-flips when both goat doors are available; (b) Monty Fall — opened a random door of the other two and happened to reveal a goat; (c) Monty Crawl — always opens the lowest-numbered door he legally can.

Solution. Prior $\frac{1}{3}$ each. The likelihood of "opens door 3" per car position $(1, 2, 3)$: (a) standard — $(\frac{1}{2}, 1, 0)$, so the posterior on door 2 is $\frac{1}{1+1/2} = 2/3$; (b) Fall — $(\frac{1}{2}, \frac{1}{2}, 0)$, posterior $1/2$: the accidental reveal carries less information because it was equally cheap either way; (c) Crawl — he opens door 2 whenever it is legal, so opening door 3 forces the car onto door 2: switching wins with certainty. Same revealed fact, three answers — the likelihood belongs to the *action under the protocol*, not to the fact.

Chapter 7

Logarithms

Practice problems

Problem 7.1. How many doublings take 1 to 512 — that is, what is $\log_2 512$?

Solution. $512 = 2^9$, so the counter above 512 reads 9: nine doublings. The answer is read off the exponent — no calculator, and no float log call, which is exactly how the series computes every integer log.

Problem 7.2. A student writes $\log_{10}(100+1000) = \log_{10} 100+\log_{10} 1000 = 5$. What is the true value, roughly — and why is the base-2 near-miss $\log_2(2+2) = 2$ no excuse?

Solution. $100 + 1000 = 1100$, and $\log_{10} 1100 = 3 + \log_{10} 1.1 \approx 3.0414$ — between 3 and 4, nowhere near 5. The product law converts multiplication, not addition: $\log_{10}(100\times 1000)$ *is* 5. The base-2 check succeeds only because $2 + 2$ happens to equal 2×2 — a coincidence of the number 2, not a law.

Problem 7.3. A fair coin comes up heads ten times running, probability $(1/2)^{10} = 1/1024$. What is $-\log_2$ of that probability? And what is the pH-style reading of a concentration of 10^{-7}?

Solution. $1/1024$ is ten halvings of 1, so $-\log_2 \frac{1}{1024} = 10$. Likewise 10^{-7} is seven tenfoldings down: $-\log_{10} 10^{-7} = 7$ — a pH of 7. Small probabilities are not scary in log space; they are just step counts with a minus sign.

Problem 7.4. A test has likelihood ratio 9 per head (and 1/9 per tail). Starting from even odds, the sequence H, H, H, T arrives. Where does the marker sit on the base-3 evidence ruler, and what are the odds?

Solution. On the ruler: $0 \to 2 \to 4 \to 6 \to 4$. Position 4 in base 3 means odds $3^4 = 81{:}1$. Multiplicatively: $9 \times 9 \times 9 \times \frac{1}{9} = 81$ — the same answer, because the ruler and the waterfall are the same data. One tail undoes exactly one head, by construction.

Problem 7.5. [stretch] A model scores 324 frames at probability 0.1 each. In float64, what does the product return, and what does the counter row say? Then price a log-space *addition*: two merging paths each carry weight 2^{-10} — what is $\log_2$ of their sum, by the chapter's closing identity?

Solution. The product is exactly 0.0 — 0.1^{324} underflows float64's floor — while the counter row records $324 \times (-1) = -324$ without complaint: the very numbers the underflow scene puts on screen. For the addition, apply the identity in base 2 with $a = b = 2^{-10}$:

$$\log_2(a+b) = -10 + \log_2\!\left(1 + 2^{\,-10-(-10)}\right) = -10 + \log_2 2 = -9,$$

and indeed $2^{-10} + 2^{-10} = 2^{-9}$ exactly. A trellis α-add costs one shifted log — no detour through value space, no cliff.

Chapter 8

e and the natural logarithm

Practice problems

Problem 8.1. Compute the split-year table's first two refinements exactly, as fractions: $(1+\frac{1}{2})^2$ and $(1+\frac{1}{4})^4$. Verify that the second exceeds the first and that both sit below 3.

Solution. $(3/2)^2 = 9/4 = 2.25$ and $(5/4)^4 = 625/256 \approx 2.4414$. Since $625/256 > 9/4$ (cross-multiply: $625 \cdot 4 = 2500 > 9 \cdot 256 = 2304$) and $625/256 < 3$ ($625 < 768$), the table climbs while staying under the ceiling — the two wrong intuitions already in trouble by $n = 4$.

Problem 8.2. Read the slope of 2^x at 0 the series' way: compute $(2^{dt} - 1)/dt$ at $dt = 0.1$, 0.01 and 0.001. What is the ratio settling toward?

Solution. The one-sided ratios are 0.7177, 0.6956 and 0.69339 — each closer to $\ln 2 = 0.6931\ldots$ than the last. The readout settles; the settling value is base 2's stride length in natural units, though at this point in the story it is still a mystery constant.

Problem 8.3. Base 8's mystery constant is exactly three of base 2's, because $8 = 2^3$. Without any new measurement, what is base 32's constant?

Solution. $32 = 2^5$ is five base-2 strides, so its constant is $5 \times \ln 2 \approx 5 \times 0.6931 \approx 3.4657$ — and indeed $\ln 32 = 3.4657\ldots$ The constants obeyed the strip's laws before they were named; naming them $\ln b$ changed nothing they do.

Problem 8.4. An account grows continuously at 2% per year. How long until it doubles? What would the popular rule of 72 say, and which answer is the mathematics?

Solution. $t = \ln 2/r = 0.6931\ldots/0.02 \approx 34.66$ years. The rule of 72 says $72/2 = 36$ years. The mathematics is $\ln 2$; 72 is a divisibility convention — easier to split by $2, 3, 4, 6, 8$ — that trades accuracy for mental arithmetic.

Problem 8.5. [stretch] Compute $\ln(e^{-1000} + e^{-1001})$. What happens on the naive route through value space, and what does the repaid identity give?

Solution. Naively, e^{-1000} underflows to exactly 0.0 in float64, so the route through value space asks for $\ln 0$ — the cliff. The identity never leaves log space: with $\ln a = -1000$ and $\ln b = -1001$,

$$\ln(a + b) = -1000 + \ln\bigl(1 + e^{-1}\bigr) = -1000 + \mathrm{log1p}(e^{-1}) \approx -999.6867.$$

Cross-check by factoring the *smaller* term instead: $-1001 + \ln(1 + e)$ — the same number, as it must be. This is the logarithms chapter's cliff walked in the repaid notation: every symbol now earns its keep.

Chapter 9

Random variables

Practice problems

Problem 9.1. Three fair flips, X = number of heads. Stamp the eight cells and sort: what is the pmf of X?

Solution. The eight sequences stamp as one 0, three 1's, three 2's and one 3, so the pmf is $(1,3,3,1)/8$ — and the counts are $\binom{3}{k}$, the counting chapter's H/T words one flip early. Four unequal bars from eight equal cells: the equiprobability bias refuted at $n = 3$.

Problem 9.2. Where does the fulcrum balance for a fair die? And for the biased die with double weight on the six (weights $1, 1, 1, 1, 1, 2$ over 7)?

Solution. Fair: $E = (1 + \cdots + 6)/6 = 21/6 = 7/2 = 3.5$ — not a face. Biased: $E = (1 + 2 + 3 + 4 + 5 + 2 \cdot 6)/7 = 27/7 \approx 3.857$ — also not a face, and *moved*: the balance point belongs to the measure, not to the faces.

Problem 9.3. Two fair dice, S = their sum. Compute $E[S]$ twice: once from the distribution of S, once with no distribution at all.

Solution. By the distribution: the diagonal counts over 36 give $E[S] = \sum_{s=2}^{12} s \cdot P(S = s) = 252/36 = 7$. By linearity: $E[S] = E[X_1] + E[X_2] = 3.5 + 3.5 = 7$, using only that both dice are fair — the second route never asks whether the dice are independent, because it does not need to know.

Problem 9.4. Assemble the binomial pmf for $n = 4$, $p = 1/4$ the sorted-square way: cell counts times one cell's area. Then check $E = np$ by the definition sum.

Solution. Column k weighs $\binom{4}{k}(1/4)^k(3/4)^{4-k}$:

$$\frac{81}{256}, \frac{108}{256}, \frac{54}{256}, \frac{12}{256}, \frac{1}{256},$$

summing to $256/256 = 1$. Each one-head cell is $(1/4)(3/4)^3 = 27/256$, and there are $\binom{4}{1} = 4$ of them — coefficient as cell count, power as one cell's area. The definition sum gives $E = (0 \cdot 81 + 1 \cdot 108 + 2 \cdot 54 + 3 \cdot 12 + 4 \cdot 1)/256 = 256/256 = 1$; indicators agree instantly: $np = 4 \times 1/4 = 1$.

Problem 9.5. [stretch] Zero successes in n trials of chance $1/n$: compute $(1-1/10)^{10}$ exactly and $(1-1/100)^{100}$ to six places, and compare both with $1/e$. From which side, and in what order, do they approach it?

Solution. $(9/10)^{10} = 3486784401/10^{10} = 0.3486784401$ exactly, and $(99/100)^{100} \approx 0.366032$, against $1/e \approx 0.367879$. Both sit *below* $1/e$ and climb toward it — $0.348678 < 0.366032 < 0.367879$ — the split-year ceiling mirrored: $(1 + 1/n)^n$ crowds e from below, and $(1 - 1/n)^n$ crowds $1/e$ the same way. Rare events at rate one-per- trial-count leave you empty-handed about 37% of the time, almost regardless of n.

Chapter 10

Softmax and likelihood

Practice problems

Problem 10.1. Compute softmax$(2, 1, 0)$ by hand to four decimal places ($e^2 \approx 7.3891$, $e^1 \approx 2.7183$). Why do your three printed values not sum to 1.0000 — and is that a defect?

Solution. The denominator is $e^2 + e^1 + e^0 \approx 7.3891 + 2.7183 + 1 = 11.1073$, giving $(7.3891,\ 2.7183,\ 1)/11.1073 = (0.6652,\ 0.2447,\ 0.0900)$. The printed values sum to 0.9999: each was rounded down by a little under half a unit in the last place, and the three roundings happen to accumulate. The exact shares sum to 1 — the display is honest about being a display, and a scene (or a homework answer) must never show the rounded values totalling 1.0000 digit by digit.

Problem 10.2. The naive normalizer $z/\sum_j z_j$ also produces numbers summing to 1. Evaluate it on $z = (2, 1, 0)$ and on the shifted $z' = (3, 2, 1)$, and name both failures. Then show in one line why softmax gives *exactly* the same distribution on z and z'.

Solution. Naively, $z = (2, 1, 0)$ gives $(2/3,\ 1/3,\ 0)$ — the score-0 class is erased, where softmax kept 0.0900 for it — and $z' = (3, 2, 1)$ gives $(1/2,\ 1/3,\ 1/6)$: the shares moved under a shift that changed nothing about the scores' order or gaps. (With a negative score the recipe can even go negative.) For softmax, $e^{z_i+c} = e^c e^{z_i}$, so

$$\frac{e^{z_i+c}}{\sum_j e^{z_j+c}} = \frac{e^c\, e^{z_i}}{e^c \sum_j e^{z_j}} = \frac{e^{z_i}}{\sum_j e^{z_j}} :$$

the common factor cancels against the normalizer — shift invariance, the property that forced the exponential in the first place.

Problem 10.3. Run the softmax recipe in base 2 on $z = (2, 1, 0)$: compute $2^{z_i} / \sum_j 2^{z_j}$ exactly. At what temperature T does ordinary (base-e) softmax reproduce your answer, and does the winner change at any base $b > 1$?

Solution. $2^2 + 2^1 + 2^0 = 7$, so the shares are exactly $(4/7,\ 2/7,\ 1/7)$ — a rational softmax, against the irrational base-e values. Since $b^z = e^{z \ln b}$, base b is base e at temperature $T = 1/\ln b$; here $T = 1/\ln 2 \approx 1.4427$ — base 2 is just a warmer dial. The winner never changes for any $b > 1$ (or any $T > 0$): exponentiation in any base above 1 is monotone, so the largest score keeps the largest share.

Problem 10.4. For the coin data $k = 3$ heads in $n = 4$ flips (count lens, $L(p) = 4p^3(1 - p)$): evaluate L at $p = 1/4$, $1/2$, and $3/4$; verify that the peak sits at $\hat{p} = k/n$; and compute the exact area under L over $[0, 1]$. What does that area prove?

Solution. $L(1/4) = 4 \cdot \frac{1}{64} \cdot \frac{3}{4} = \frac{3}{64} \approx 0.0469$; $L(1/2) = 4 \cdot \frac{1}{8} \cdot \frac{1}{2} = \frac{1}{4}$; $L(3/4) = 4 \cdot \frac{27}{64} \cdot \frac{1}{4} = \frac{27}{64} \approx 0.4219$ — the largest of the three, and a sweep (or the next chapter's derivative) confirms the maximum is exactly at $\hat{p} = 3/4 = k/n$, the observed proportion. The area: $\int_0^1 (4p^3 - 4p^4)\, dp = 1 - \frac{4}{5} = \frac{1}{5} \neq 1$. So the likelihood, swept over p, is not a probability distribution over the parameter — each *column* of the two-lens table normalizes over outcomes, but nothing ever normalized the row. A prior and a renormalization would turn it into one; that is the Bayes chapter's move, not this one's.

Problem 10.5. [stretch] On the three-frame matrix of the closing section, compute the exact probability of the path A, A, ε and the sum of its logs. Then the survival argument: how many frames of per-frame probability 0.1 does it take to kill a float32 product outright, and what does the sum of logs read at that point?

Solution. $0.7 \times 0.6 \times 0.7 = 147/500 = 0.294$ exactly; the logs $\ln 0.7 + \ln 0.6 + \ln 0.7 \approx -0.3567 - 0.5108 - 0.3567 = -1.2242$, and $e^{-1.2242} \approx 0.294$ round-trips. At three frames both routes are fine — the contrast lives further out: 0.1^{45} still stores as float32's smallest subnormal, and the forty-sixth factor lands *exactly* 0.0 — the product is gone, and no later factor can bring it back. The sum of logs reads $46 \ln 0.1 = -105.9189$, an unremarkable number an accumulator carries without complaint. A real utterance has hundreds of frames; this is why the loss lives in log space, not merely why it is convenient there.

Chapter 11

The derivative toolkit

Practice problems

Problem 11.1. Write the forward quotient of $f(x) = x^2$ at $x = 1$ for a step h, simplify it exactly, and evaluate it at $h = 1, \frac{1}{10}, \frac{1}{100}$. What does the simplified form say "settling" means?

Solution. $\big((1+h)^2-1\big)/h = (2h+h^2)/h = 2+h$ exactly — no limit machinery needed to see it. The values are 3, 2.1, 2.01: each entry *is* $2+h$, so settling is nothing more than the step fading out of the answer, leaving the slope 2. (A curiosity worth one aside: the symmetric quotient $\big(f(1+h) - f(1-h)\big)/2h$ equals exactly 2 for every h — a special property of quadratics, which is why the honest settling story is told with forward quotients.)

Problem 11.2. Differentiate $y = (2x)^2$ at $x = 1$ by the chain rule, naming both rates and where each is evaluated. A classmate answers 4 — reconstruct their mistake, and refute it with the forward quotients at $h = \frac{1}{10}$ and $\frac{1}{100}$.

Solution. Inner map $u = 2x$: rate 2. Outer map $y = u^2$: rate $2u$ evaluated at the inner *output* $u = 2$, so 4. The chain multiplies: $2 \times 4 = 8$. The classmate evaluated the outer rate at the input $x = 1$, getting $2 \times 2 = 4$. The quotients decide it: $y = 4x^2$, so $\big(4(1+h)^2 - 4\big)/h = 8 + 4h$, reading 8.4 and 8.04 — headed for 8, not 4.

Problem 11.3. Derive $\frac{d}{dx} \ln x = 1/x$ from $\frac{d}{dx} e^x = e^x$ using the undo identity $e^{\ln x} = x$, and check it numerically at $x = 2$ with the forward quotient at $h = \frac{1}{100}$.

Solution. Differentiating $e^{\ln x} = x$: the left side, by the chain rule, is $e^{\ln x} \cdot (\ln x)'$, and the right side is 1. But $e^{\ln x} = x$, so $x \cdot (\ln x)' = 1$ and $(\ln x)' = 1/x$ — the undo-never-cancel flip carrying its own derivative. At $x = 2$: $(\ln 2.01 - \ln 2)/0.01 = 0.4988$, already close to $1/2$ and settling as h shrinks (the quotients run 0.4879, 0.4988, 0.4999 $\to$ 0.5 at $h = 10^{-1}, 10^{-2}, 10^{-3}$).

Problem 11.4. For the likelihood $L(p) = 4p^3(1-p)$: write $\ln L$, differentiate it term by term into the score, evaluate the score at $p = 1/2$, 0.7, and 0.8, and solve the zero. Then find the general $\hat{p}$ for k heads in n flips.

Solution. $\ln L = \ln 4 + 3\ln p + \ln(1-p)$, so the score is $3/p - 1/(1-p)$ (the constant contributes nothing; $\ln(1-p)$ differentiates to $-1/(1-p)$ by the chain rule). Values: $6 - 2 = +4$ at $p = 1/2$; $3/0.7 - 1/0.3 = 30/7 - 10/3 = 20/21 \approx +0.9524$ at 0.7; $15/4 - 5 = -5/4 = -1.25$ at 0.8. The sign flips across $p = 3/4$, and the zero is linear: $3/p = 1/(1-p) \Rightarrow 3(1-p) = p \Rightarrow \hat{p} = 3/4$. In general the score is $k/p - (n-k)/(1-p)$, whose zero $k(1-p) = (n-k)p$ gives $\hat{p} = k/n$ — the observed proportion, for every k and n at once.

Problem 11.5. [stretch] With the correct class scoring 2 in $z = (2, 1, 0)$: differentiate $\mathrm{NLL}(z) = \mathrm{LSE}(z) - z_a$ component by component, and evaluate the gradient to four decimal places. Then let the correct class's score fall: on $z = (2, 1, t)$ with the truth in the third slot, compute the slope $\partial\,\mathrm{NLL}/\partial t$ at $t = 0, -2, -5$. What is the slope pressing toward, and why can it never arrive?

Solution. Each component of $\nabla\mathrm{LSE}$ is a softmax share, and subtracting z_a subtracts 1 from the truth's component: $\nabla\,\mathrm{NLL} = p - \text{one-hot} = (0.6652 - 1,\ 0.2447,\ 0.0900) = (-0.3348,\ 0.2447,\ 0.0900)$, whose exact components sum to 0 (the shares sum to 1, and one 1 was removed). On $z = (2, 1, t)$ the slope in t is $p_c - 1$ where $p_c = \mathrm{softmax}(z)_3$: at $t = 0, -2, -5$ it reads -0.9100, -0.9868, -0.9993. The walk presses toward -1 but never arrives: p_c is a ratio of positive exponentials, so it is positive at every finite t — the loss becomes as straight as you like, with slope -1 only as a limit. (At $t = -10$ the share is $\approx 4.5 \times 10^{-6}$; write it in scientific notation — rounding the slope to -1.0000 would claim the limit was reached.)

Chapter 12

Gradient descent

Practice problems

Problem 12.1. On $L(w) = w^2$ with $\eta = \frac{1}{4}$ and $w_0 = 4$, write out w_1, w_2, w_3 exactly. After how many steps does $|w_k|$ first drop below 0.01?

Solution. The update is $w \leftarrow w - \frac{1}{4} \cdot 2w = \frac{w}{2}$, so $w_1 = 2$, $w_2 = 1$, $w_3 = \frac{1}{2}$, and in general $w_k = 4 \cdot 2^{-k}$. Requiring $4 \cdot 2^{-k} < 0.01$ means $2^k > 400$, first true at $k = 9$ (where $w_9 = 4/512 \approx 0.0078$). Nine steps — and each one cost a single multiplication, because on this bowl the rule *is* a fixed shrink factor.

Problem 12.2. Still on $L(w) = w^2$ from $w_0 = 4$: for $\eta \in \{\frac{1}{4}, \frac{3}{4}, 1, \frac{5}{4}\}$, find the per-step factor and classify each walk (converges, oscillates and converges, oscillates forever, diverges). For which η does the walk reach the bottom at all?

Solution. The factors are $\frac{1}{2}$, $-\frac{1}{2}$, -1, $-\frac{3}{2}$. With $|1 - 2\eta| < 1$ the distance to the bottom shrinks geometrically: $\eta = \frac{1}{4}$ glides in, and $\eta = \frac{3}{4}$ crosses the bottom every step but still converges, because overshooting is fine so long as each landing is *closer*. At $\eta = 1$ the factor is -1: the walk ping-pongs $4, -4, 4, \ldots$ forever. At $\eta = \frac{5}{4}$ each step lands one-and-a-half times farther out: divergence. The walk converges exactly for $0 < \eta < 1$ — on this bowl; the threshold is set by the bowl's curvature, which is why one landscape's safe rate is another's cliff.

Problem 12.3. On the double well $L(w) = \frac{w^4}{4} - \frac{w^2}{2}$ with $\eta = 0.1$: where does the walk settle from $w_0 = 0.5$, from $w_0 = -0.5$, and from $w_0 = 0$? Classify each stopping place with the sign-change test.

Solution. From 0.5 the walk settles at 1; by symmetry, from -0.5 it settles at -1. Around $w = 1$ the slope crosses $-$ to $+$: a valley, and likewise at -1. From $w_0 = 0$ the walk never moves — $L'(0) = 0$, so the update is exactly zero — yet around 0 the slope crosses $+$ to $-$: a hilltop. All three are legitimate stopping places of the same rule. Descent stops at flat ground; only the sign change says which kind of flat ground it was.

Problem 12.4. [stretch] From the walk in the chapter — loss 0.7181 at the start, 0.1602 after 10 steps, 0.0088 after 200, 0.0003 after 5000 — compute the average per-step shrink factor of the loss over the first ten steps, and over the last 4800. What do the two factors say about where a training run spends its time?

Solution. First leg: $(0.1602/0.7181)^{1/10} \approx 0.86$ — the loss loses about 14% of itself *every step*. Last leg: $(0.0003/0.0088)^{1/4800} \approx 0.9993$ — under a tenth of a percent per step. The two factors differ by two orders of magnitude in effect, which is the front-loading of the chapter in one line: early on the gradient is large and each step earns; near the bottom the gradient (and so the step) has shrunk, and equal-looking spans of the loss cost geometrically more steps. Reading a training curve means reading its slope — the long flat tail is the rule working, not failing. (The answer script re-runs the full five-thousand-step descent from scratch and reproduces all four anchored losses to four decimal places.)

Chapter 13

The CTC gradient: softmax minus occupancy

Practice problems

Problem 13.1. At unit weights, run the backward recurrence on the AB trellis ($T = 4$, states $\varepsilon, A, \varepsilon, B, \varepsilon$): $\beta_4 = 1$ at the two accepting states, 0 elsewhere. Compute the $t = 1$ column of suffix counts, and check its two initial states against the flagship path count.

Solution. From $\beta_4 = (0, 0, 0, 1, 1)$: the $t = 3$ column is $(0, 1, 1, 2, 1)$ — e.g. state B has stay 1 + advance 1; the A state's skip to B is legal since $A \neq B$. Then $t = 2$ gives $(1, 4, 3, 3, 1)$ — $\beta_2(A) = \text{stay } 1 + \text{advance } 1 + \text{skip } 2 = 4$ — and $t = 1$ gives $(5, 10, 6, 4, 1)$. The two initial states (leading blank and first letter): $5 + 10 = 15$, the same count the forward sweep found from the other end.

Problem 13.2. Using the forward counts $(1, 1, 0, 0, 0)$, $(1, 2, 1, 1, 0)$, $(1, 3, 3, 4, 1)$, $(1, 4, 6, 10, 5)$ and your backward counts, form the per-cell products $\alpha\beta$ at every cell and sum each column. What do the four sums say, and why must they agree?

Solution. The product columns are $(5, 10, 0, 0, 0)$, $(1, 8, 3, 3, 0)$, $(0, 3, 3, 8, 1)$ and $(0, 0, 0, 10, 5)$ — each summing to 15. $\alpha_t(s)\beta_t(s)$ counts the paths *through* cell (t, s) — ways in times ways out — and since every one of the 15 paths occupies exactly one state at every frame, summing over a column counts every path exactly once, whichever column you pick. (The 8 at (t=2, A) is the waist: 2 prefixes $\times$ 4 suffixes.) On the real matrix the same sweep returns $P(\text{AB} \mid X) = 0.4877$ four times — the constant column, with probabilities riding the counts.

Problem 13.3. Under uniform outputs ($y_t(k) = 1/3$ for all t, k): compute $P(\text{AB} \mid X)$, the occupancy column at $t = 1$, and the gradient $y - \gamma$ there. What is remarkable about what the computation never used?

Solution. Each of the 15 paths weighs $(1/3)^4 = 1/81$, so $P = 15/81 = 5/27$. With all weights equal, occupancy is path counts over 15: at $t = 1$, ten paths start in A and five in the leading blank, so $\gamma_1 = (2/3, 0, 1/3)$ across (A, B, ε). The gradient is $y - \gamma = (1/3 - 2/3,\ 1/3 - 0,\ 1/3 - 1/3) = (-1/3,\ +1/3,\ 0)$. The input never entered: at an uninformative start, the error signal is determined by the target sequence's topology alone — which is exactly the diffuse first stage of training, and the seed of blank's topological head start at larger T.

Problem 13.4. On the worked matrix, frame 1 has $y_1 = (0.7, 0.2, 0.1)$ and occupancy $\gamma_1 = (0.9028, 0, 0.0972)$. Write the gradient column, verify it sums to zero, and explain why B's entry equals its softmax output exactly — no rounding involved.

Solution. $y - \gamma = (0.7 - 0.9028,\ 0.2 - 0,\ 0.1 - 0.0972) = (-0.2028,\ +0.2000,\ +0.0028)$, summing to 0: both y and γ sum to 1 over the classes, so their difference sums to zero — mass pushed off the truth's cells must land on the others. At frame 1 a path for AB can occupy only the leading blank or the first A; the B state is unreachable, so its occupancy is exactly 0 and its gradient is exactly $y = +0.2000$ — the loss pushes a provably useless activation straight down. (The same mechanism, run against the wrong transcript BA, flips frame 1's A to exactly $+0.7000$.)

Problem 13.5. Derive the identity. Starting from $\partial L/\partial \ln y_t(k) = -\gamma_t(k)$ and the log-softmax derivative $\partial \ln y_j/\partial u_k = \delta_{jk} - y_k$, show $\partial L/\partial u_t(k) = y_t(k) - \gamma_t(k)$, naming the step where the constant column does the work.

Solution. Chaining,

$$\frac{\partial L}{\partial u_t(k)} = \sum_j \frac{\partial L}{\partial \ln y_t(j)} \frac{\partial \ln y_t(j)}{\partial u_t(k)} = -\sum_j \gamma_t(j)\big(\delta_{jk} - y_t(k)\big).$$

The δ term picks out $-\gamma_t(k)$; the second term is $+\,y_t(k)\sum_j \gamma_t(j)$, and the sum is 1 — the column checksum, the same constant column that verified the bookkeeping, now killing the Jacobian's second term. What remains is $y_t(k) - \gamma_t(k)$: softmax output minus occupancy. With a single surviving path, γ is one-hot and the identity is the derivative chapter's $p -$ one-hot — the general case received the special one.

Problem 13.6. [stretch] Count label cells over all AB-paths — with no input anywhere — at $T = 4$ and $T = 5$: how many of the path-cells carry A, B, and blank? At which T does blank take the lead, and what does that say about the first gradient step from an uninformative start?

Solution. At $T = 4$: 60 path-cells split $(A, B, \varepsilon) = (21, 21, 18)$ — the labels lead, and blank is *not* dominant; any unconditioned "blank always dominates" claim dies on this case. At $T = 5$: 175 path-cells split $(56, 56, 63)$ — blank ahead, and it stays ahead for every larger T, with its share growing toward 2/3 in the one-letter limit under uniform outputs. Since uniform-init occupancy equals these count shares, the very first descent step already pushes blankward once $T \geq 5$ — the topological half of peakiness, before weight sharing adds the trap.

Chapter 14

Decoding: from scores to a transcript

Practice problems

Problem 14.1. Best-path decode this three-frame model over $\{A, B, \varepsilon\}$: $y_1 = (0.5, 0.2, 0.3)$, $y_2 = (0.4, 0.3, 0.3)$, $y_3 = (0.2, 0.3, 0.5)$ (listed as A, B, ε). Then decide: does the full sum over paths crown the same transcript here?

Solution. The frame favourites are A, A, ε, collapsing to A. The brute-force posterior (all 27 paths, grouped by transcript) agrees: A leads with mass 0.3170, ahead of the empty transcript and of every two-letter candidate. Greedy is not always wrong — with one frame-wise favourite chain clearly ahead, path mass concentrates and the max speaks for the sum. The failures live where frames hedge.

Problem 14.2. Generalise the chapter's construction: two frames, $y_t(\varepsilon) = q$, $y_t(A) = 1-q$. For which q does greedy decode the empty transcript while the sum prefers A? Locate the exact boundary.

Solution. Greedy takes the blank whenever $q > \frac{1}{2}$. The empty transcript truly wins only when $q^2 > 1 - q^2$, i.e. $q > 1/\sqrt{2} \approx 0.7071$. So on the whole band $\frac{1}{2} < q < 1/\sqrt{2}$ greedy is wrong — at $q = 0.6$ the chapter's 0.36 versus 0.64, and even at $q = 0.7$ the sum still prefers A (0.49 versus 0.51). The blank must win each frame by a $\sqrt{2}$ margin, not a majority, before "nothing was said" is the honest answer.

Problem 14.3. Run the two-ledger beam by hand on the chapter's construction ($y_t(A) = 0.4$, $y_t(\varepsilon) = 0.6$, $T = 2$, width 2), tracking (p_b, p_{nb}) for every prefix at each frame. Confirm the final masses against the alignment chapter's sums.

Solution. After frame one: ' ' $(0.6, 0)$ and A $(0, 0.4)$. Frame two: the empty prefix keeps $0.6 \times 0.6 = 0.36$ in p_b; A collects the new letter $0.6 \times 0.4 = 0.24$ and the silent merge $0.4 \times 0.4 = 0.16$ into $p_{nb} = 0.40$, and its blank extension $0.4 \times 0.6 = 0.24$ in p_b. Totals: 0.36 versus 0.64 — identical to the brute-force sums, as the committed script asserts: at this size the beam prunes nothing and *is* the forward recurrence.

Problem 14.4. [stretch] Uniform coin frames: $y_t(A) = y_t(\varepsilon) = 0.5$ over $T = 3$. Compute the true $P(\text{AA})$, then show what a one-ledger beam — a single mass per prefix, so every letter extension is treated as a new letter — would report for AA, and diagnose the error.

Solution. Only $A\varepsilon A$ collapses to AA, so $P(\text{AA}) = 0.5^3 = 0.125$. The one-ledger beam, unable to tell merged repeats from blank-separated ones, extends the whole of prefix A's mass by A into AA at every step and reports 0.375 — a three-fold overcount, crediting AA with the mass of $AA\varepsilon$ and εAA, both of which actually collapse to A. The two ledgers are not an optimisation; they are the collapse map's grammar carried into the search, and dropping either one changes the answers, not just the speed.